Final Project Report/Video

Graded

2 Days, 17 Hours Late

Group

Sabrina Nicacio
Enrico Vittori
Colin Skinner
View or edit group

Total Points

120 / 120 pts

Question 1

Project Title

0 / 0 pts

✔ **+ 0 pts** Correct

Question 2

Author Information **0** / 0 pts

2.1 **Author 1 Stanford Email (e.g., "suid@stanford.edu")** **0** / 0 pts

✔ **+ 0 pts** Correct

2.2 **Author 2 Stanford Email (e.g., "suid@stanford.edu")** **0** / 0 pts

✔ **+ 0 pts** Correct

2.3 **Author 3 Stanford Email (e.g., "suid@stanford.edu")** **0** / 0 pts

✔ **+ 0 pts** Correct

2.4 **Author 4 Stanford Email (e.g., "suid@stanford.edu")** **0** / 0 pts

✔ **+ 0 pts** Correct

Question 3

Permission to Share on Course Website

0 / 0 pts

+ 0 pts Correct

✔ **+ 0 pts** Incorrect

Question 4

Peer Review Exclusion

0 / 0 pts

+ 0 pts Correct

✔ **+ 0 pts** Incorrect

Question 5

File Upload

120 / 120 pts

✔ **+ 10 pts** **Abstract or Overview**
There is a reasonable abstract, overview or table of contents present in the paper/presentation. May be written or spoken.

✔ **+ 20 pts** **Submitted**

✔ **+ 10 pts** **Problem Clearly Stated**
Succinct and clear problem description appropriate for other members of the course.

✔ **+ 10 pts** **Approach Clearly Stated**
Succinct and clear approach description appropriate for other members of the course.

✔ **+ 10 pts** **Appropriateness of Approach**
It is justified why the specific approach in question was chosen for the problem at hand.

✔ **+ 10 pts** **Structure of Paper / Presentation**
The paper/presentation is divided into sections that make it easy to read/listen and follow.

✔ **+ 10 pts** **Analysis and Results**
The approach described was quantitatively analyzed according to a set of clearly described metrics, and the results of this analysis presented as figures/tables/graphs/etc.

✔ **+ 10 pts** **Adequacy of References and Discussion of Prior Work**
If the project is a follow-on from an ongoing larger research project, previous or closely related publications are summarized and cited in order to provide adequate background.
If the project is new but closely related to existing research in literature, the relevant sources are cited and discussed.
Finally, citations are provided for any algorithms and software not developed in the scope of the project.

✔ **+ 10 pts** **Quality of Discussion and Conclusion**
The conclusions follow logically from the presented results. Salient aspects of the experiments that were introduced earlier are included in the discussion.

✔ **+ 10 pts** **Clarity of Drawings, Graphs, Tables**
The drawings, graphs and tables included support the points that are made in the paper/presentation. Complex concepts that require diagrammatic or visual presentation are appropriately presented.

✔ **+ 10 pts** **Writing Quality / Presentation Quality**
Terms are properly defined and there is no unnecessary jargon. Sentences come to logical conclusions. The writing/speaking structure does not interfere with comprehension.

– 20 pts Not a sequential decision making problem

– 0.5 pts Need more clarity in figure/table

– 5 pts Deductions for formatting/writing

+ 0 pts No submission

– 5 pts Insufficient written work for sections

– 5 pts Insufficient discussion

– 5 pts Insufficient quantitative analysis/experimentation

– 5 pts Insufficient connection to prior work and the literature

– 5 pts Chosen approach does not align with the intended problem formulation or stated objectives

– 20 pts Not a Sequential Decision Making Problem

– 5 pts The presentation of the approach and the work lacked sufficient clarity and depth.

– **5 pts** Underspecified details of the sequential decision making problem

– **3 pts** Minor details missing

– **5 pts** No abstract or overview (either written or spoken).

Fantastic work!! Very impressive project.

Question 6

Group Contributions

0 / 0 pts

✔ **+ 0 pts** Included here

+ 0 pts Included in report/video

+ 0 pts Individual

+ 0 pts Not included

Q1 Project Title
0 Points

POMDP Framework for Orbital Rendezvous and Docking

Q2 Author Information

0 Points

Leave empty if not applicable

Q2.1 Author 1 Stanford Email (e.g., "suid@stanford.edu")

0 Points

snicacio@stanford.edu

Q2.2 Author 2 Stanford Email (e.g., "suid@stanford.edu")

0 Points

colin.skinner@stanford.edu

Q2.3 Author 3 Stanford Email (e.g., "suid@stanford.edu")

0 Points

evittori@stanford.edu

Q2.4 Author 4 Stanford Email (e.g., "suid@stanford.edu")

0 Points

N.A.

Q3 Permission to Share on Course Website

0 Points

I am ok with our report/video being shared publicly on the course website for future students taking AA228/CS238.

- (●) Yes
- ( ) No

Q4 Peer Review Exclusion
0 Points

Are you opting out of the peer review? **Only select yes if approved by the course staff!** This option is only for students who work on confidential projects. Opting out of the peer review only means your paper will not be part of the peer review process, but you still have review two other projects!

- ○ Yes
- ◉ No

Q5 File Upload

120 Points

Upload your project report as .pdf file or the video as .mp4 file.

AA228_Final_Project.pdf Download

Your browser does not support PDF previews. You can download the file instead.

Q6 Group Contributions
0 Points

If you submitted a video and are working in a group, please use the text box below to summarize each group member's contributions.

Sabrina developed the full spacecraft dynamics and state-transition model, shaped the state-estimation setup, and built the end-to-end simulation by writing the complete systems-integration code that connected the dynamics, EKF, and POMDP modules. This included implementing the POMDP reward function and transition logic. She also built the results analysis notebook, generating plots and performance metrics, and structured the simulation so the team could tune parameters without modifying core code.
Colin contributed to state estimation and results sections of both the code and report as his base contribution. His extra 30 hours were spent heavily streamlining/tuning the simulation, generating the figures, data, and analysis for the report, and writing all other sections (prev. work, state estimation, results, introduction, conclusion).
Enrico wrote the POMDP and MCTS model in the code, and the decision making, problem formulation, and results sections of the report.